Completing the square

1 Halve the coefficient of x and then square it.

2

a No

b 9

3 $\frac{9}{4}$

4

a 25

b 1

c 8

d $\frac{1}{4}$

e $\left(\frac{m}{2}\right)^2$

f $\frac{4}{25}$

5

a $x^2 + 4x + 4 = (x + 2)^2$

b $x^2 - 5x + \frac{25}{4} = \left(x - \frac{5}{2}\right)^2$

c $x^2 - \frac{7}{4}x + \frac{49}{64} = (x - \frac{7}{8})^2$

6

a $(x + 6)^2 - 36$

b $(x - 6)^2 - 36$

c $(x + 1)^2 + 1$

d $(x - 7)^2 + 7$

e $(x + 2)^2 - 2$

f $(x - 3)^2 - 4$

g $\left(x + \frac{3}{2}\right)^2 + \frac{15}{4}$

h $\left(x - \frac{7}{2}\right)^2 + \frac{7}{4}$

7

a $3\left(x + \frac{3}{2}\right)^2 + \frac{5}{4}$

b $4\left(x - \frac{11}{8}\right)^2 - \frac{9}{16}$

8 $4\left(\left(x - \frac{9}{2}\right)^2 - \frac{1}{2}\right)$

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9

a $(b+7)^2$ **b** $(y-5)^2$ **c** $(x+5)^2$ **d** $(x+7)^2$

e $(x-6)^2$ **f** $(x-8)^2$ **g** $(9x-4)^2$ **h** $(9x+2)^2$

i $(7x-2)^2$ **j** $(9+x)^2$ **k** $(6-x)^2$ **l** $(5+6r)^2$

m $7(d+8)^2$ **n** $4(x+5)^2$ **o** $3(x-4y)^2$ **p** $-6(r+8)^2$

q $-3(x-2)^2$ **r** $\left(b-\frac{7}{2}\right)^2$ **s** $\left(x+\frac{3}{2}\right)^2$ **t** $\left(x-\frac{3}{2}\right)^2$

10

a $y=\left(x-3+\sqrt{2}\right)\left(x-3-\sqrt{2}\right)$ **b** $y=\left(x+4+\sqrt{2}\right)\left(x+4-\sqrt{2}\right)$

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